Sufficiency

1. Let $X_1, \ldots, X_n|\theta \overset{\text{iid}}{\sim} N(\theta, \sigma^2)$ where σ^2 is known. Show that $\sum_{i=1}^n X_i$ and/or $\bar{X}$ is sufficient for θ. *Hint: expand the quadratic.*

2. Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Unif}[0, \theta]$ for $\theta > 0$. Find a sufficient statistic for θ. *Hint: indicators?*

3. Let $X_1, \ldots, X_n|\theta \overset{\text{iid}}{\sim} \text{Poisson}(\theta)$. Demonstrate that $T(\mathbf{X}) = \sum_{i=1}^n X_i$ is sufficient for θ using the definition of sufficiency.

4. Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} f(x|\theta)$, where

$$f(x|\theta) = \begin{cases} \frac{\theta}{(1+x)^{\theta+1}} & x \geq 0 \\ 0 & o.w., \end{cases}$$

for $\theta > 0$. Find a sufficient statistic for θ. Can you also find a more convenient sufficient statistics that's represented as a sum? *Hint: manipulate/massage the expression around!*